



Q.1 (B)

$$\frac{T_{r+1}}{T_r} = \frac{1}{3}$$

$$\text{If } T_1 = a \quad ; \quad T_2 = \frac{a}{3} \quad ; \quad T_3 = \frac{a}{3^2}, \dots$$

$$\therefore T_7 = \frac{a}{3^6} = \frac{1}{243} \Rightarrow \frac{a}{3^5 \times 3} = \frac{1}{3^5} \Rightarrow a = 3$$

$$\therefore T_r \cdot T_{r+1} = \frac{a}{3^{r-1}} \times \frac{a}{3^r} = \frac{a^2}{3^{2r-1}} = \frac{3^2}{3^{2r-1}}$$

$$\sum_{r=1}^{\infty} T_r \cdot T_{r+1} = \left(3 + \frac{1}{3} + \frac{1}{3^3} + \frac{1}{3^5} + \dots \right) = \frac{3}{1 - \frac{1}{3^2}} = \frac{27}{8}$$

Q.2 (A)

$$S_n = \sum_{r=1}^n \left(\frac{1}{6^n} {}^n C_r \left(\sum_{t=0}^{r-1} {}^r C_t 4^t \right) \right)$$

$$= \sum_{r=1}^n \frac{1}{6^n} {}^n C_r [(1+4)^r - 4^r]$$

$$= \frac{1}{6^n} \sum_{r=1}^n ({}^n C_r 5^r - {}^n C_r 4^r)$$

$$= \frac{1}{6^n} [(1+5)^n - 1 - ((1+4)^n - 1)] = \frac{6^n - 5^n}{5^n} = 1 - \left(\frac{5}{6}\right)^n$$

$$\sum_{r=1}^{\infty} 1 - S_n = \sum_{n=1}^{\infty} \left(\frac{5}{6}\right)^n = \frac{\frac{5}{6}}{1 - \frac{5}{6}} = 5. \text{ Ans.}$$

Q.3 (C)

$$x_1 \quad 1 \quad x_2 \quad 2 \quad x_3 \quad 3 \quad x_4 \quad 4 \quad x_5$$

where x_1 denotes no. of 7's

$$x_1 + x_2 + x_3 + x_4 + x_5 = 10$$

where $x_2, x_3, x_4 \geq 2$

and $x_1, x_5 \geq 0$

using beggar method



Ans: ${}^8C_4 \times 4! = 14 \times 5!$

Q.4 (B)

Any factor of this number should be of the form $2^a \cdot 3^b \cdot 5^c$.

For the factor to be a perfect square a, b, c have to be even.

a can take values 0, 2, 4. b can take values 0, 2 and c can take values 0, 2.

Total number of perfect squares = $3 \cdot 2 \cdot 2 = 12$ **Ans.**

$$25 \times 36 \times 52 = (5)^2 \times (6)^2 \times (2)^2 \times 13 = (2)^4 \times (3)^2 \times (5)^2 \times (13)^1$$

Q.5 (D)

$$A + B + C = 19$$

Case I : If A = 0 then number of ways = ${}^{20}C_1$

Case II : If A = 2 then number of ways = ${}^{18}C_1$

Case III : If A = 3 then number of ways = ${}^{16}C_1$

⋮ ⋮ ⋮ ⋮

⋮ ⋮ ⋮ ⋮

Case X : If A = 18 then number of ways = 2C_1

$$\therefore \text{total no. of ways} = {}^{20}C_1 + {}^{18}C_1 + {}^{16}C_1 + \dots + {}^2C_1 = 110$$

Q.6 (C)

Case A: There is only one 9, one 8 and one 7 in the number.

Hence there has to be one digit from {1,2,3,4,5,6} repeated thrice.

Total number of ways in which such a number can exist :

$$= {}^6C_1 \cdot \frac{5!}{3! \cdot 2!} = 60 = n(A)$$

Case B: One of the three digit {9,8,7} is repeated thrice.

Hence there will be one digit from {1,2,3,4,5,6}.

Total number of ways in which such a number can exist:

$$\Rightarrow {}^3C_1 \cdot {}^6C_1 \cdot \frac{5!}{2! \cdot 2!} = 540$$

Total possible telephone numbers :

$$= 60 + 540 = 600$$

Probability = $60/600 = 1/10$ **Ans.**

Q.7 (B)

Let the number on the ball picked first = a, second = b and third = c

The three numbers a, b and c are distinct. Hence for favorable cases we have to select 3 numbers out of 10 and numbers will arrange in the required order in only one way hence number of favorable ways will be ${}^{10}C_3 = 120$

$$\text{Required probability} = \frac{{}^{10}C_3}{10 \times 10 \times 10} = \frac{3}{25} \text{ **Ans.**}$$



Q.8 (D)

[Sol. 287/prob/SC]

$$E_1 : \text{FANAND} \xrightarrow{\frac{1}{2}} P\left(\frac{A}{E_1}\right) \rightarrow \frac{2}{4}$$

$$E_2 : \text{FAFANAND} \xrightarrow{\frac{1}{2}} P\left(\frac{A}{E_2}\right) \rightarrow \frac{4}{6}$$

Required probability = $\frac{\frac{1}{2} \times \frac{2}{4}}{\frac{1}{2} \times \frac{2}{4} + \frac{1}{2} \times \frac{4}{6}} = \frac{3}{7}$ Ans.

Q.9 (C)

$$P(A) = \frac{3}{4}; \quad P(B) = \frac{1}{2}; \quad P(C) = \frac{5}{8}$$

$$\therefore \begin{array}{c} \swarrow \quad \downarrow \quad \searrow \\ AB\bar{C} \quad A\bar{B}\bar{C} \quad \bar{A}B\bar{C} \end{array}$$

$$\Rightarrow \text{Probability} = \left(\frac{3}{4} \times \frac{1}{2} \times \frac{3}{8}\right) + \left(\frac{3}{4} \times \frac{1}{2} \times \frac{3}{8}\right) + \left(\frac{1}{4} \times \frac{1}{2} \times \frac{3}{8}\right) = \frac{21}{64} \text{ Ans.}$$

Q.10 (C)

Events (independent)

$$\begin{array}{c} \swarrow \quad \searrow \\ A \quad B \end{array}$$

$$\Rightarrow P(A \cap B) = P(A) P(B)$$

$$\text{Now, } P(A) + P(B) - 2(A \cap B) = \frac{26}{49} \quad \dots\dots(1)$$

$$\text{Also, } P(A) + P(B) - P(A) \cdot P(B) = \frac{34}{49} \quad \dots\dots(2)$$

$\therefore$ on solving, we get

$$P(A) = \frac{2}{7}, \quad P(B) = \frac{4}{7} \text{ Ans.}$$

Q.11 (C)

Q.12 (C)

Q.13 (AD)

Q.14 (C)

Maximum sum of 7 digit number can be 63. For sum be 61, we have to divide from seven nines.

So, either 9, 9, 9, 9, 9, 9, 7 or 9, 9, 9, 9, 9, 8, 8.

$$\text{Let } x = a_1 + a_3 + a_5 + a_7$$

$$y = a_2 + a_4 + a_6$$

For number to be divisible by 11,



$x - y = 11n, n \in W$ as $x > y$ (Think?)
where $x \leq 36 ; y \leq 27$

$x - y$	$x + y$	x	y	
$x - y = 0$	$x + y = 61$	30.5	30.5	Not possible
$x - y = 11$	$x + y = 61$	36	25	Only case possible
$x - y = 22$	$x + y = 61$	41.5	19.5	Not possible
$x - y = 33$	$x + y = 61$	47	14	Not possible
$x - y = 44$	$x + y = 61$	8.5	8.5	Not possible
$x - y = 55$	$x + y = 61$	58	3	Not possible

$$\therefore x = 36 ; y = 25$$

as $x = 36$

$$\therefore a_1 = a_3 = a_5 = a_7 = 9$$

a_2, a_4, a_6 has to be from 9, 8, 8 or 9, 9, 7

Hence number of such number will be 6. **Ans.**

Q.15 [4]

$$f(f(x)) = f(x) + f^2(x) + f^4(x) + f^8(x) + \dots$$

Coefficient of x^{10} in $f(x) = 0$

Coefficient of x^{10} in $f^2(x) = 2$

Coefficient of x^{10} in $f^4(x) = {}^4C_2 + {}^4C_1 = 10$

Coefficient of x^{10} in $f^8(x) = {}^8C_2 = 28$

Q.16 [0141]

$$\left(x + \frac{1}{x} + 1\right)^6 = \sum_{r=1}^6 {}^6C_r \left(x + \frac{1}{x}\right)^r$$

$$\therefore a_0 = 1 + {}^6C_2 \cdot {}^2C_1 + {}^6C_4 \cdot {}^4C_2 + {}^6C_6 \cdot {}^6C_3$$

$$= 1 + 30 + 90 + 20 = 141. \text{ Ans.}$$

Q.17 [7]

Select 3 places for A, B, C in 7C_3 way $\times \times \times \times \times \times \times$
and arrange them only in 1 way.

Remaining 4 places can be arranged in $4!$ ways

$$\therefore \text{Total ways} = {}^7C_3 \cdot 4! = 35 \cdot 24 = 840.$$

$$\Rightarrow N = 840$$

$$\therefore \left(\frac{N}{120}\right) = \frac{840}{120} = 7 \text{ Ans.}$$

Q.18 [0006]

Case-1: Consider the numbers between 550 – 599

4, 5, 6, 7, 8, 9

1st place in one way (i.e. 5)

2nd place in 5 ways (i.e. 5, 6, 7, 8, 9)

3rd place in 3 ways (i.e. 5, 7, 9)

Number of numbers in this case = $1 \cdot 5 \cdot 3 = 15$.

fixed

5		
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Case-2: 600 – 799 (both inclusive)

Number of numbers = $2 \cdot 6 \cdot = 36$

Total = $15 + 36 = 51$.

Number of digits in $N = 6$. **Ans.**

Q.19 [0013]

$$\begin{aligned}\sum_{r=0}^9 \frac{{}^{11}C_r \times {}^{10}C_{r+1}}{r+1} &= \sum_{r=0}^9 \frac{r+1}{12} \frac{{}^{12}C_{r+1} \cdot {}^{10}C_{r+1}}{r+1} \\&= \frac{1}{12} [{}^{12}C_1 {}^{10}C_1 + {}^{12}C_2 {}^{10}C_2 + \dots + {}^{12}C_{10} {}^{10}C_{10}] \\&= \frac{1}{12} [{}^{12}C_1 {}^{10}C_9 + {}^{12}C_2 {}^{10}C_8 + \dots + {}^{12}C_{10} {}^{10}C_0] \\&= \frac{1}{12} [{}^{22}C_{10} - 1] = \left[\frac{{}^{22}C_{10} - 1}{12} \right]\end{aligned}$$

Q.20 [0023]

Last toss can be $\begin{cases} \text{Head then first } (n-2) \text{ should not yield 2 consecutive head} \\ \text{Tail then first } (n-1) \text{ should not yield 2 consecutive head} \end{cases}$

$P(H) = 3x$, $P(T) = x$

Now, $3x + x = 1 \Rightarrow x = \frac{1}{4}$

$\therefore P(H) = \frac{3}{4}$, $P(T) = \frac{1}{4}$

$\therefore P_n = P_{n-2} \times \frac{3}{4} \times \frac{1}{4} + \frac{1}{4} P_{n-1}$

$16P_n = 3P_{n-2} + 4P_{n-1}$